

LABORATORY RECORD

Course:	Full Stack Web Development
Course Code:	23CM4121
Week No.:	2
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Date:	30-08-2026

1. TITLE

Design Webpage Using HTML5 Canvas & CSS3 Styling

2. AIM

To construct a styled webpage using HTML5 semantic elements, 2D Canvas graphics rendering, and CSS3 layout styles.

3. OBJECTIVE

- Implement CSS3 linear gradient backgrounds and flexbox containers.
- Style card components with border-radius, box-shadows, and hover animations.
- Draw 2D shapes (rectangle, circle, line) and text using HTML5 Canvas API.

4. THEORY

CSS3 enhances web aesthetics with linear gradients, flexible box layouts, box shadows, and transition animations. HTML5 Canvas provides an interactive scriptable drawing context for graphics rendering.

5. ALGORITHM / PROCEDURE

1. Create html&css.html file.
2. Define styles for linear gradient background, header, and flexbox card grid.
3. Create canvas section and initialize 2D drawing context.
4. Draw rectangle, circle, line, and name text using canvas JavaScript commands.
5. Add styled content cards for HTML5, CSS3, and JavaScript.
6. Verify visual styling and graphics in browser.

6. PROGRAM

html&css.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>

  <style>
    /* General Styling */
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background: linear-gradient(to right, #74ebd5, #9face6);
    }

    /* Header */
    header {
      background: #2c3e50;
      color: white;
      text-align: center;
      padding: 20px;
      font-size: 32px;
      text-shadow: 2px 2px 5px rgb(222, 3, 3);
    }

    /* Navigation */
    nav {
      background: #34495e;
      padding: 15px;
      text-align: center;
    }

    nav a {
      color: white;
      text-decoration: none;
      margin: 20px;
      font-size: 18px;
      padding: 8px 15px;
      transition: 0.3s;
      border-radius: 5px;
    }

    nav a:hover {
      background: white;
      color: #34495e;
    }

    /* Canvas Section */
    .canvas-section {
      text-align: center;
      margin: 30px;
    }

    canvas {
      border: 3px solid black;
      background: white;
      box-shadow: 0px 0px 15px gray;
    }

    /* Content Cards */
    .container {
      display: flex;
```

```
        justify-content: center;
        gap: 25px;
        margin: 30px;
        flex-wrap: wrap;
    }

    .card {
        width: 280px;
        background: white;
        border: 2px solid #555;
        border-radius: 15px;
        padding: 20px;
        margin: 10px;
        box-shadow: 5px 5px 15px gray;
        text-align: center;
    }

    .card h2 {
        font-family: Georgia, serif;
        color: darkblue;
        text-shadow: 1px 1px 2px gray;
    }

    .card p {
        font-family: Verdana, sans-serif;
        font-size: 16px;
    }

    /* Button */
    .btn {
        background: #3498db;
        color: white;
        padding: 10px 20px;
        border: none;
        border-radius: 8px;
        cursor: pointer;
        transition: 0.3s;
    }

    .btn:hover {
        background: green;
        transform: scale(1.1);
    }

    /* Footer */
    footer {
        background: #2c3e50;
        color: white;
        text-align: center;
        padding: 15px;
        margin-top: 20px;
    }
}
</style>
</head>

<body>

    <!-- Header -->
    <header>
        My Styled Webpage
    </header>

    <!-- Navigation -->
    <nav>
        <a href="#">Home</a>
```

```
<a href="#">About</a>
<a href="#">Services</a>
<a href="#">Contact</a>
</nav>

<!-- Canvas -->
<div class="canvas-section">
  <h2>HTML5 Canvas</h2>

  <canvas id="myCanvas" width="600" height="300">
    Your browser does not support Canvas.
  </canvas>
</div>

<!-- Content Cards -->
<div class="container">

  <div class="card">
    <h2>HTML5</h2>
    <p>
      HTML5 provides the structure of web pages with modern elements
      like canvas, audio, video, and semantic tags.
    </p>
    <button class="btn">Learn More</button>
  </div>

  <div class="card">
    <h2>CSS3</h2>
    <p>
      CSS3 makes websites attractive using gradients, shadows,
      animations, hover effects, and responsive layouts.
    </p>
    <button class="btn">Explore</button>
  </div>

  <div class="card">
    <h2>JavaScript</h2>
    <p>
      JavaScript adds interactivity to web pages and controls HTML5
      Canvas for drawing graphics.
    </p>
    <button class="btn">Read More</button>
  </div>

</div>

<!-- Footer -->
<footer>
  © 2026 My Styled Webpage | Designed by Bhanu Prasad
</footer>

<!-- Canvas Script -->
<script>
  var canvas = document.getElementById("myCanvas");
  var ctx = canvas.getContext("2d");

  // Rectangle
  ctx.fillStyle = "red";
  ctx.fillRect(40, 40, 120, 80);

  // Circle
  ctx.beginPath();
  ctx.arc(280, 80, 50, 0, 2 * Math.PI);
  ctx.fillStyle = "blue";
```

```
ctx.fill();

// Line
ctx.beginPath();
ctx.moveTo(400, 40);
ctx.lineTo(550, 150);
ctx.strokeStyle = "green";
ctx.lineWidth = 4;
ctx.stroke();

// Name Text
ctx.font = "28px Arial";
ctx.fillStyle = "black";
ctx.fillText("Bhanu Prasad", 180, 250);
</script>
```

```
</body>
```

```
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
background: linear-gradient(...)	Applies smooth multi-color background gradient
display: flex; justify-content: center	Centers card elements horizontally with flexbox
canvas.getContext('2d')	Provides 2D drawing surface for canvas graphics
ctx.fillRect(), ctx.arc(), ctx.stroke()	Renders geometric shapes directly into canvas

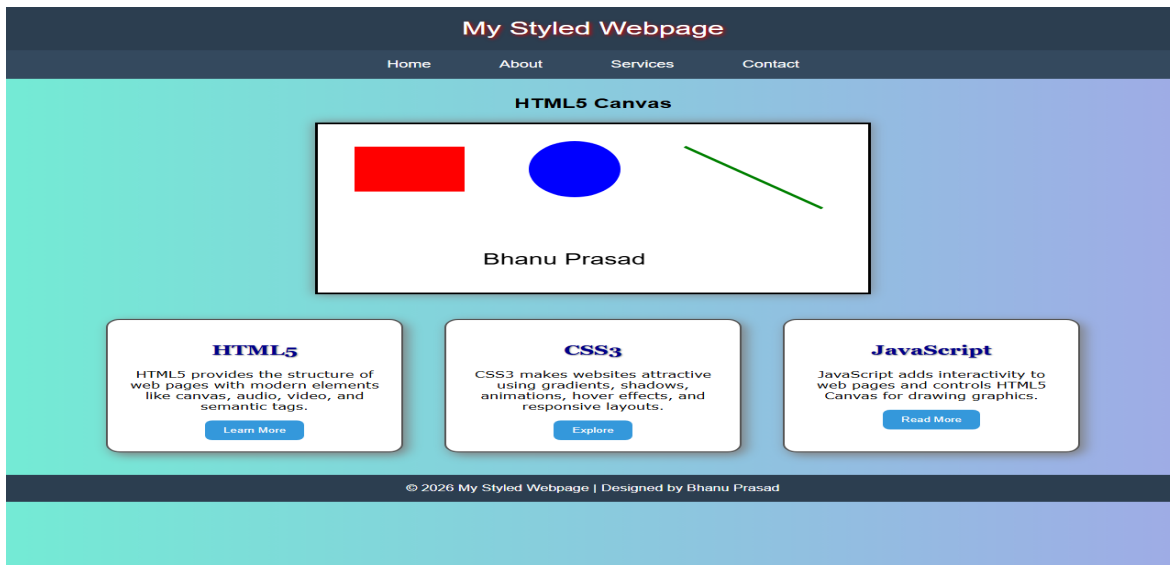
8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Page Styling	Open in web browser	Layout renders cards and gradients cleanly	Pass
TC-02	Canvas Graphics	Check canvas element	Canvas graphics and text render accurately	Pass

9. OUTPUT

Output 1: Execution Screenshot



10. RESULT

Thus, the experiment for Week 2 (Design Webpage Using HTML5 Canvas & CSS3 Styling) was successfully executed and verified.